

Ashwin Krishnamurthy

Mechatronics engineer building the future of space and defense tech.

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RESEARCH EXPERIENCE

Rutgers University, Materials Science & Engineering Dec 2024 – Present

Research Intern, microMechanics of Deformation Group · Piscataway, NJ

- Formulated theory on the relationship between disconnections, grain boundary misorientation, and migration.
- Developed script to identify and classify defect content within 7,304 minimum-energy aluminum structures.

NASA STEM Enhancement in Earth Science (SEES) May 2025 – Dec 2025

Research Intern, selected from a national pool with a ~5% acceptance rate

- Determined the distance and age of open cluster NGC 2194 through photometric analysis.
- Presented poster results at the AGU annual meeting and the MIT Undergraduate Research Technology Conference.

New Jersey Astronomical Association May 2024 – Present

Qualified Observer

- Certified to operate the observatory's 26-inch telescope for public programs and research.
- Led outreach booths at public star parties, recruiting 100+ new observatory members.

PROJECTS & LEADERSHIP

NASA App Development Challenge, Co-Captain Sep 2024 – May 2025

- National winner for an Unreal Engine 5 simulation of the Artemis II mission trajectory with key flight data.
- Earned a fully funded team trip to NASA Johnson Space Center, presenting to NASA executives.
- Invited to speak at the HELIX Discovery Showcase opening Rutgers' HELIX research building.

CubeSat, MIT Beaver Works Summer Institute, Co-Captain Dec 2024 – Aug 2025

- Placed 4th nationally in the BWSI Build-a-CubeSat Challenge with a wildfire-detection satellite that ran a VLA model trained on NASA GOES data.

FIRST Tech Challenge Team 24536, Mechanical Lead Oct 2023 – Apr 2025

- Designed competition mechanisms and integrated the full sensor suite, earning the Innovate Award.

Math League, Edison Academy Magnet School, Executive Board Aug 2023 – Jun 2026

- Secured sponsorships and authored problems for MCAMC, raised \$3,000+ for the team's ARML trip, and served as Lead Coach at Central Jersey Math Circle, the first middle school math circle in central New Jersey.

EDUCATION

Carnegie Mellon University, Pittsburgh, PA Class of 2030

B.S. Mechanical Engineering, planned minor in Robotics

Edison Academy Magnet School, Edison, NJ Class of 2026

Mechanical/Civil Engineering Track · Weighted GPA 4.61 / 4.0

AWARDS

USA Physics Olympiad Qualifier, Top 400 · 2025

3rd Place, Rutgers McLaren Symposium · 2025

NASA App Development Challenge Winner · 2025

AIME Qualifier (2×), AMC 10B Top 5% · 2022, 2023

SKILLS

Engineering: Statics, Mechanism Design, Materials Science, Heat Transfer, Manufacturing, Project Management

Software & Tools: Python, MATLAB, C++, Java, CAD, SolidWorks, Unreal Engine 5

Math & Physics: Linear Algebra, Differential Equations, Discrete Mathematics, Multivariable Calculus, Mechanics, Electricity & Magnetism, Astrophysics